

Inclusive

I value the diverse experiences and perspectives of my peers and myself.

Respectful: I listen to and respect those with different viewpoints than me.

Professional: I step back when I have shared my voice more than my classmates to make space for others to be heard.

Engaged: I step up and contribute when my voice or viewpoint has not been shared.

Supportive

I support my peers and myself through challenges to become better business leaders.

Respectful: I stay patient, helpful, and curious when my peers need extra support.

Professional: I practice leadership and allyship by encouraging others to step out of their comfort zone in class.

Engaged: I ask for clarification when I need it, knowing that others may have similar questions.

Prompt & Present

I show up to class ready to learn because I value the learning experience.

Respectful: I arrive on time so class can begin without disruptions.*

Professional: I plan ahead to only leave during breaks so my peers and professors can stay focused.

Engaged: I attend class so I can learn from and contribute to class discussions.

*Occasional late arrivals and absences should be the exception

Distraction Free

I'm mindful to minimize tech distractions and stay engaged.

Respectful: I keep my laptop and cell phone away to respect professors and speakers.**

Professional: I take notes in a way that helps other students stay focused on their learning.

Engaged: I recognize that my engagement contributes to a positive learning environment.

Academic Norms | MBA for Executives "EMBA"

**Laptops and phones may be used for online courses, instructor-approved purposes, and disability accommodations.



XMBA 200S: Data and Decisions

Lecture 1: Course Introduction

Instructor: Reed Walker

Outline

- Welcome to Data and Decisions
- Course overview and syllabus
- eBay: opportunities and pitfalls using big data

Who am I, and why am I here?

- Economist by training (Econ PhD, Columbia 2012)
 - Please call me “Reed”
- Much of (applied) economics research uses data to study questions in the social sciences
- Economics is at the frontier of statistical science
 - Important innovations in causal inference (e.g., Nobel Prize 2021, go Bears!)
 - Real world applications are paramount
 - Incentives matter in statistics!

Ground Rules and Class Norms

- Please raise hand to ask or answer a question
 - Please use mic!
- Class will start on time. Please be in seats and turn any/all devices into “Do not disturb” mode.
- Often two breaks during typical 3-hour lecture.
 - Please try to use the bathroom during these breaks and minimize disruptions during lecture.

Course introduction

- “What is Data and Decisions?”
- This is not a course that will make you a data scientist.
- This is a course that is designed to teach you:
 1. Some basic probability and statistics
 2. Frameworks or heuristics for making decisions under uncertainty
 3. Introductions to modeling / forecasting / machine learning
 4. Introductions to experimental design and implementation, vis-à-vis randomized control trials and/or AB testing

My Goals

1. To teach content/material on the previous slide
2. To foster an acumen for which tools to use when and an awareness of best practices in implementation
 - Requirement for fluency and understanding of statistics, probability, and/or data science



Which Tools to Use When

Most quantitative / data analysis can be classified either as:

1. Modeling, prediction, or forecasting problems

OR

2. Cause and effect or causal problems

Modeling, prediction, or forecasting problems

- Tools: regression models and/or machine learning

Cause and effect or causal problems

- Tools: AB testing, randomized control trials

AI Has Made This Course More Important

You were never going to run your own regressions five years post-MBA — you were going to commission them.

- Marketplace expectation is now that you have **more** fluency in this domain since execution is costless
- “can you run the procedure” no longer proxy for whether you have good judgment re: data.
- New proxy (the overarching goal of this course):
 - can you SUPERVISE an analysis you did not compute?
 - How to decide what to trust, what to ask, and what to do

Getting data analysis right

- Businesses have access to massive amounts of data
- Behavior that used to be:
 - Unobserved
 - Too dispersed
 - Too personal
- Now collected on a regular (continuous) basis:
 - Massive amounts of user data generated (e.g., online)
 - Plus the technology to store it, merge it, and use it

Many Opportunities

≡ SFGATE

[Newsletters](#)

[Account](#)

TRAVEL

Hotel booking sites caught overcharging travelers from Bay Area

In an investigative column, SFGATE contributor Keith A. Spencer exposes online price discrimination targeting San Francisco travelers

The Telegraph

Business

ALL SECT

🏠 > Business

Uber knows customers with dying batteries are more likely to accept surge pricing

By **Marion Dakers**

The car-hailing service Uber can detect when a user's smartphone is low on battery, and therefore willing to pay more to book a ride.



Getting data analysis right

Big data: are we making a big mistake?

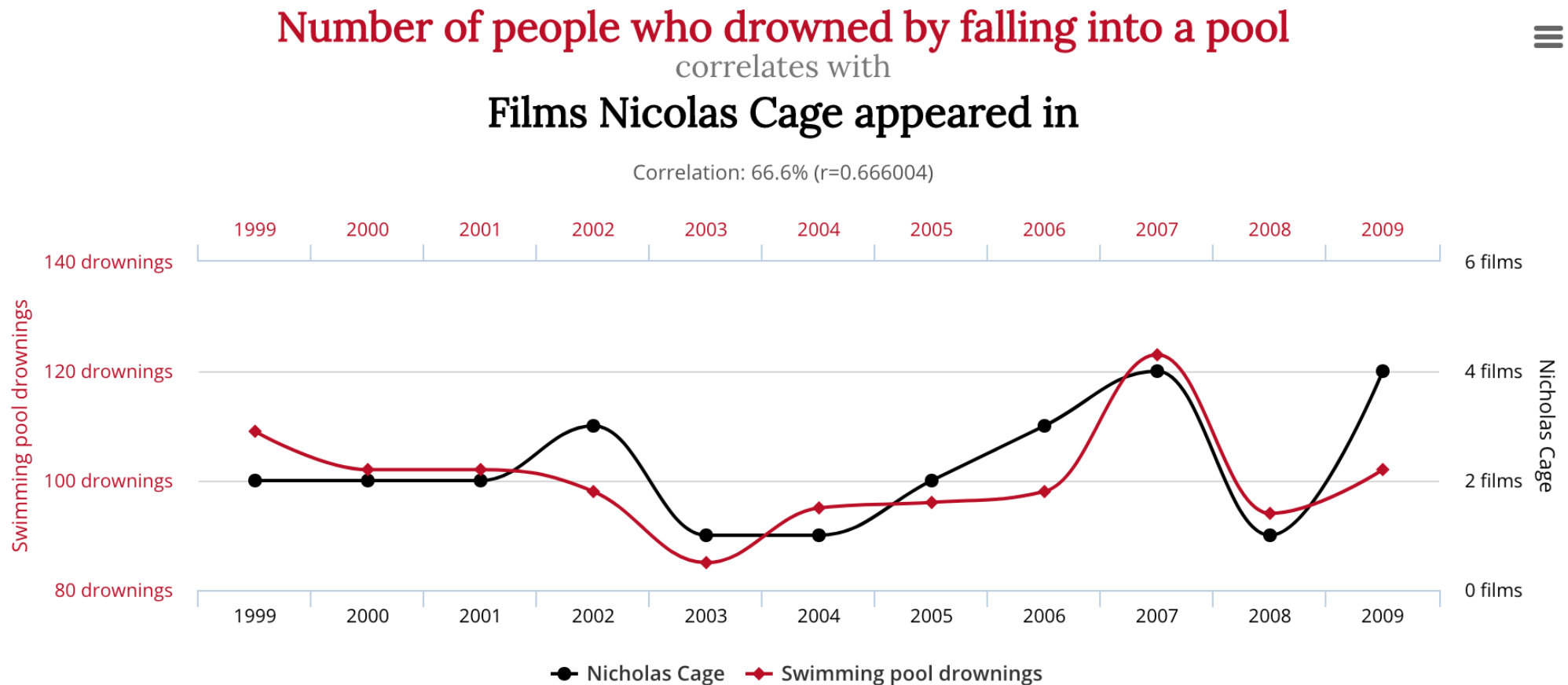
By Tim Harford

Big data is a vague term for a massive phenomenon that has rapidly become an obsession with entrepreneurs, scientists, governments and the media

“There are a lot of small data problems that occur in big data,” says Spiegelhalter.
“They don’t disappear because you’ve got lots of the stuff. They get worse.”

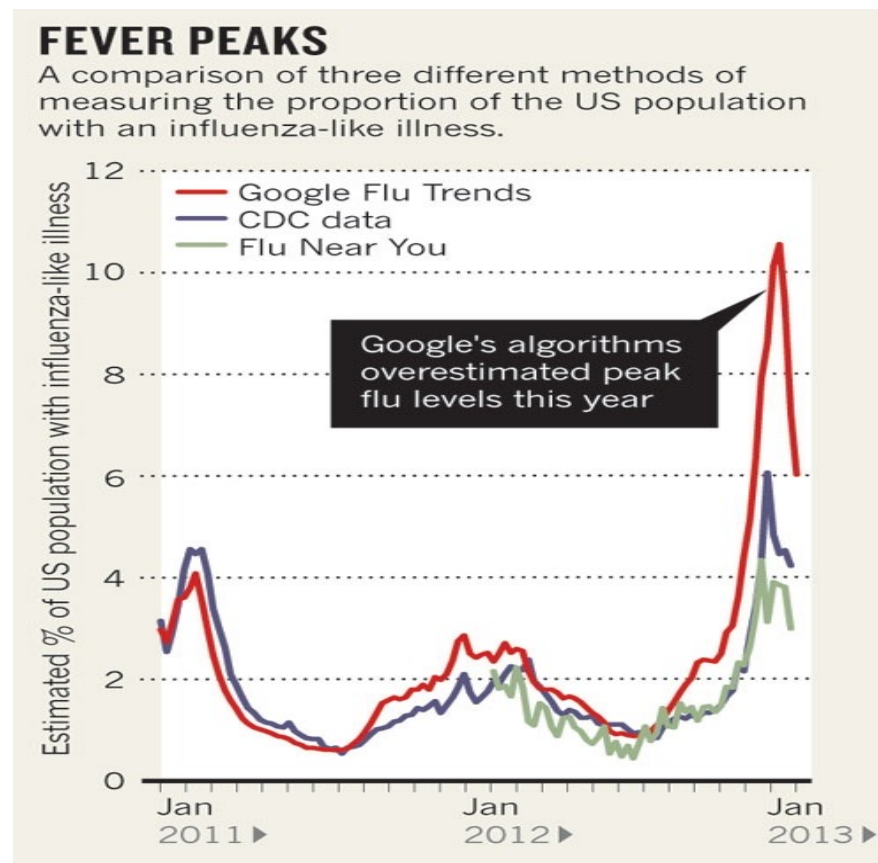
Getting data analysis right

- Possibilities for all kinds of spurious correlations



Getting data analysis right

- Powerful tools require careful interpretation and constant vigilance



Outline

- Welcome to Data and Decisions
- Course overview and syllabus
- eBay: opportunities and pitfalls using big data

Logistics

- Office hours
 - Mondays, 4-5pm via Zoom
(link on syllabus / bCourses)
 - Feel free to email me to talk at other times
- **Silvia Farina** is our GSI
 - Office hours: Tuesdays 5-6pm via Zoom
 - Email with questions at least 24hrs in advance for discussion in office hours if possible

Where to find stuff

Home

Announcements

Assignments

Quizzes

Files

Media Library
(Panopto)

Grades

People

EMBA 200S-1

Data and Decisions

Block 1

Block 2

Block 3

Block 4

Welcome

Meet Your Instructors

Lecture & Office Hours

Getting Started

Where to find stuff



Class 1 (9/10): Introduction and Samples/Surveys

Before Class

- Stine and Foster, Chapter 13
- Knowledge Check, Chapters 1-11: Covers Chapters 1-11 from pre-Term and Stine and Foster.
- Knowledge Check, Chapters 1-11: Solutions

During Class

- Lecture slides (Intro and Chapter 13)
- eBay In-Class Handout
- Sentence of the Day: "There is one population, but there are many different samples"

After Class

- Class 1 Lecture Recording

Related Articles

- [Financial incentives can improve survey representativeness](#)
 - Researchers show how financial incentives for survey respondents improves representativeness of survey, especially amongst racial minorities.

How to get questions answered

- Ask me in class.
- I will manage the conversation.
- I really welcome questions/comments from introverts!
- Come to office hours (mine and/or Silvia's)
- Come to sections.

Exams

- We have no midterm
- Final exam: Friday 11/13

Team assignments

- There will be three in class, team assignments
- There will be three, between Block, team assignments
- You can use AI, Excel, or other statistical software for these
- **Please bring at least one laptop per team this Friday and Saturday**

Assessment

- 4 main components:
 - In-class Team Exercises 10%
 - Between Block Team Exercises 40%
 - Final exam 40%
 - Class participation and attendance 10%
- Yes. It's high stakes. But so is life.
- If you have grading questions, please get in touch with me or Silvia

AI in This Course: The Rules

- Permitted and encouraged on ALL team exercises — in class and between blocks
- Use it to write code, explore data, check your reasoning, draft explanations
- You own every number. “The AI told me so” is not a defense.
- Every team submission includes a brief AI-use statement (ungraded — which tools, used for what)
- The final exam (Fri 11/13) is a paper exam: no laptops, phones, or any devices

Course overview:

Part 1: The data you want vs. the data you've got

- Managers almost never have all of the data they want . . .
 - On their customers
 - On their production facilities
 - On their employees' productivity
- If you can only collect **a sample of data**, what can you say about what is generally going on?

Questions of interest

- Suppose you want to market a new app
 - How do you learn if your (potential) customers will like it?
 - How do you learn how much your (potential) customers will pay?
- You need to sample from a population
 - How you **select your sample** matters
 - Who selects into your sample matters
- What can you say about potential **customers you will never observe**?

Course overview:

Part 2: Making decisions based on inference

- Given **limited data**, managers need to choose how to act on it
 - e.g. small pilot project that does slightly better than status quo
 - But how do you know any difference isn't due to "noise"?
 - What is the likelihood of making a mistake?
- What we need are **statistical tests** that provide us with some clarity

Questions of interest

1. Piloting a new production process in a plant:
 - Should you adopt this more broadly?
2. Your company develops a new diagnostic device
 - Does it predict disease better than existing devices?

We'll create statistical tests to better understand differences in presence of uncertainty

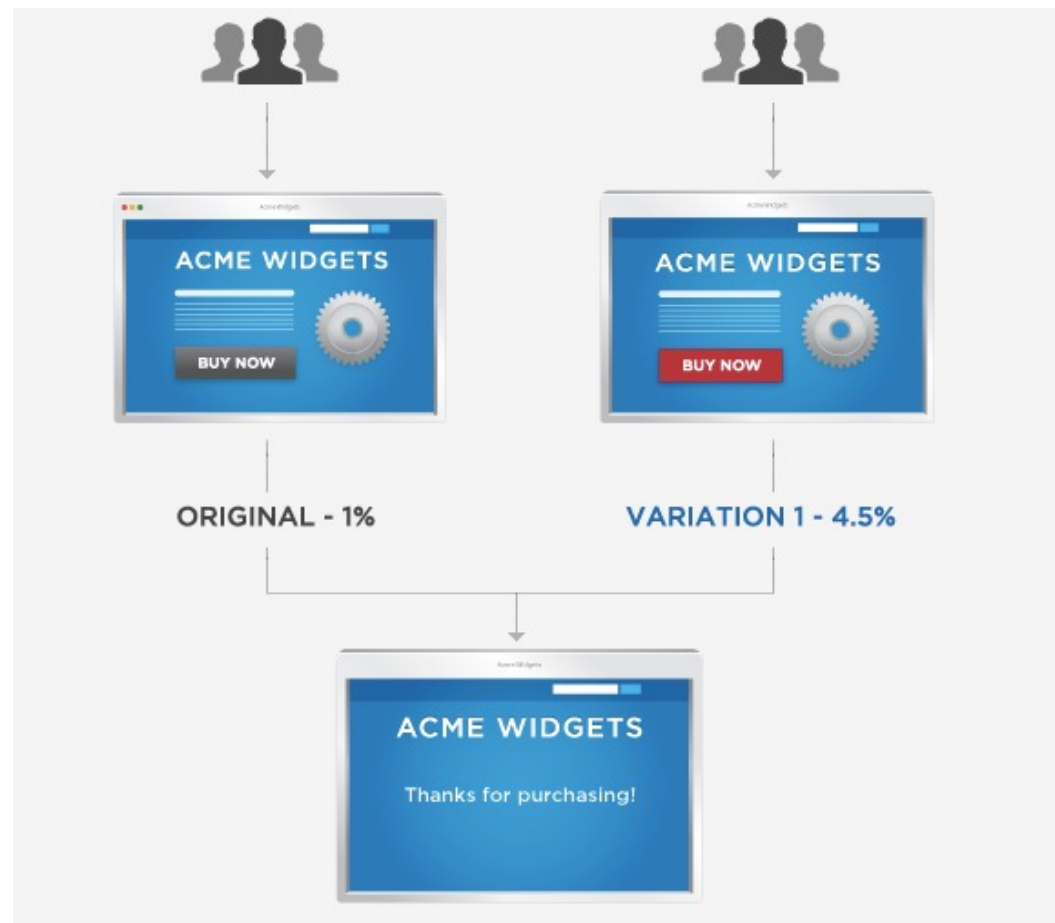
Course overview:

Part 3: Experiments and causality

- In many comparisons you make, you want to attribute differences to some **causal factor**
- Experiments are a uniquely powerful (and transparent) way of testing for causal effects
 - Non-experimental comparisons suffer from “lurking variables” or bias from omitted variables.

Questions of interest

- You can run an A/B test on all aspects of your website:



Questions of interest

- How can we improve worker effort at a firm?

Money or Monitoring: Evidence on Improving Worker Effort

Higher compensation and increased monitoring are two common strategies for addressing the moral hazard problem between firms and workers. In a field experiment with new hires at an automobile manufacturing firm in China, we randomly varied both signing bonuses and monitoring intensity. Both interventions increased worker output but through different channels: signing bonuses led to longer working hours without significant gains in performance, while enhanced monitoring improved performance as evaluated by managers. Additionally, bonuses reduced quit rates, whereas

Questions of interest

- Does Medical AI using Audio/Video “Work”



2026-08-11

Towards Expert-level Medical AI for Real-time Video Consultations

Audio-visual interaction is the standard for patient-physician consultations, enabling natural communication and effective assessment of illness through non-verbal cues. While text-based AI has shown promise, it discards essential perceptual dimensions and limits patients who cannot articulate symptoms in writing. Early efforts to extend medical AI to audio-visual interaction have demonstrated feasibility, but not reached clinician-level performance. Here, we provide the first demonstration of expert-level AI in real-time clinical video consultations using AMIE (Articulate Medical Intelligence Explorer) in a video configuration. AMIE (Video) is a Gemini-based multi-agent system integrating low-latency dialogue, clinical reasoning, and real-time audio-visual perception. To guide development, we established a taxonomy and automated evaluations for clinical audio-visual cues in telehealth settings. In a randomized Objective Structured Clinical Examination (OSCE) study with 30 primary care physicians (PCPs), 15 patient actors and 100 clinical scenarios, we compared AMIE (Video), its text-only counterpart AMIE (Text), and PCPs consulting via video. Clinical evaluators rated AMIE (Video) on par or better than PCPs

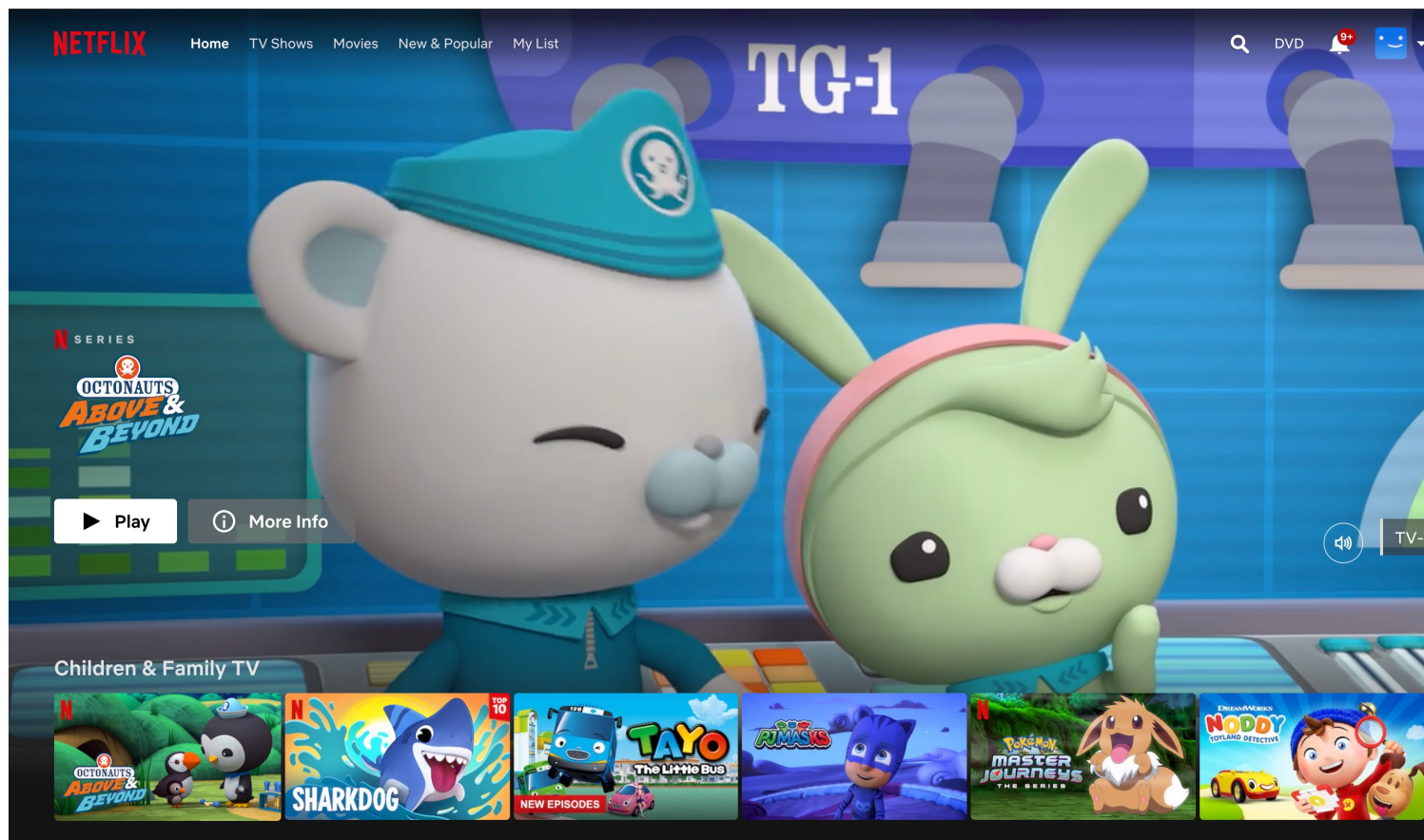
Course overview:

Part 4: Building predictive models

- You want to build a predictive model to forecast outcomes from data you have
- What's the best model you can build?

Questions of interest

- You want to recommend a new product to a customer



Questions of interest

- Risk adjustment in value-based healthcare
 - How do we determine the cost of a new patient in a capitated health care payment system?



- Credit scoring in consumer lending
 - How do we forecast the likelihood that a new applicant will repay their loan?



Outline

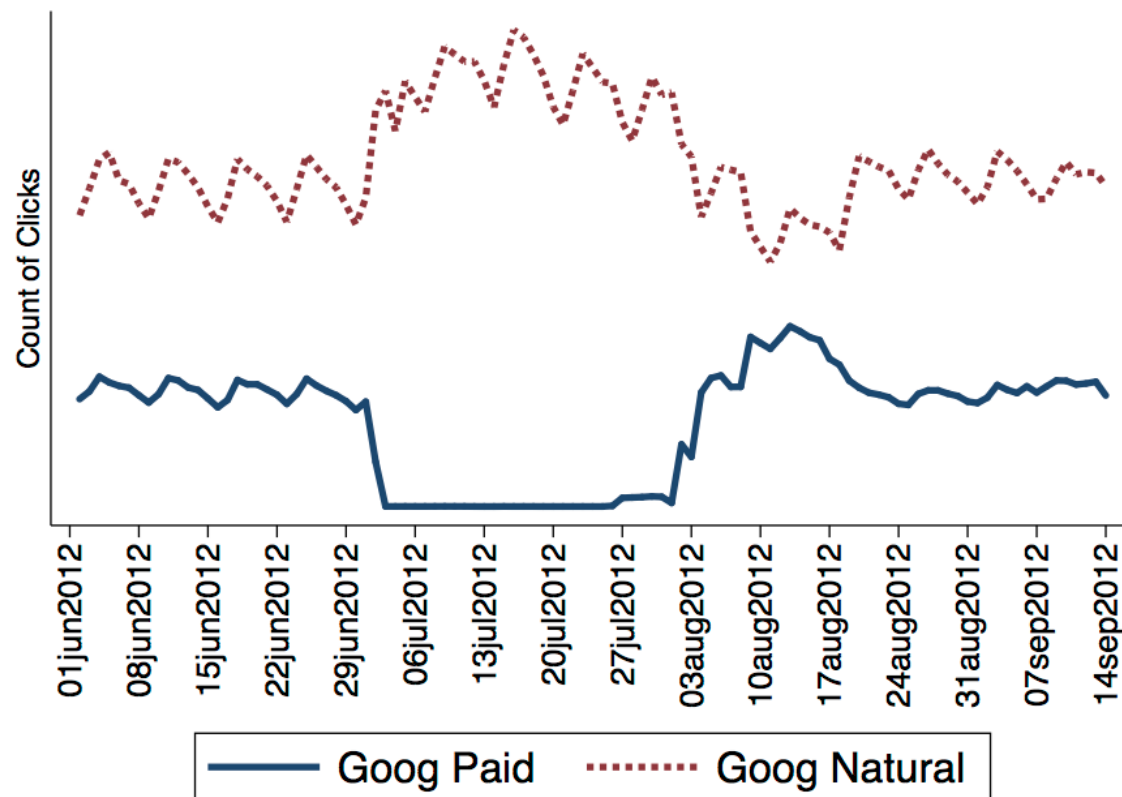
- Welcome to Data and Decisions
- Course overview and syllabus
- eBay: opportunities and pitfalls using big data

How to test whether SEM paid off?

- Getting the statistics right can save you millions
- How do we design a study to test the returns to a Search Engine Marketing (SEM) strategy?
[Ad • www.ebay.com]
- Take 10 minutes to work on the posted handout.
- Ref.: Blake, Thomas, Chris Nosko, and Steven Tadelis. 2015. "Consumer heterogeneity and paid search effectiveness: A large-scale field experiment." *Econometrica*, 83, pp.155-174.

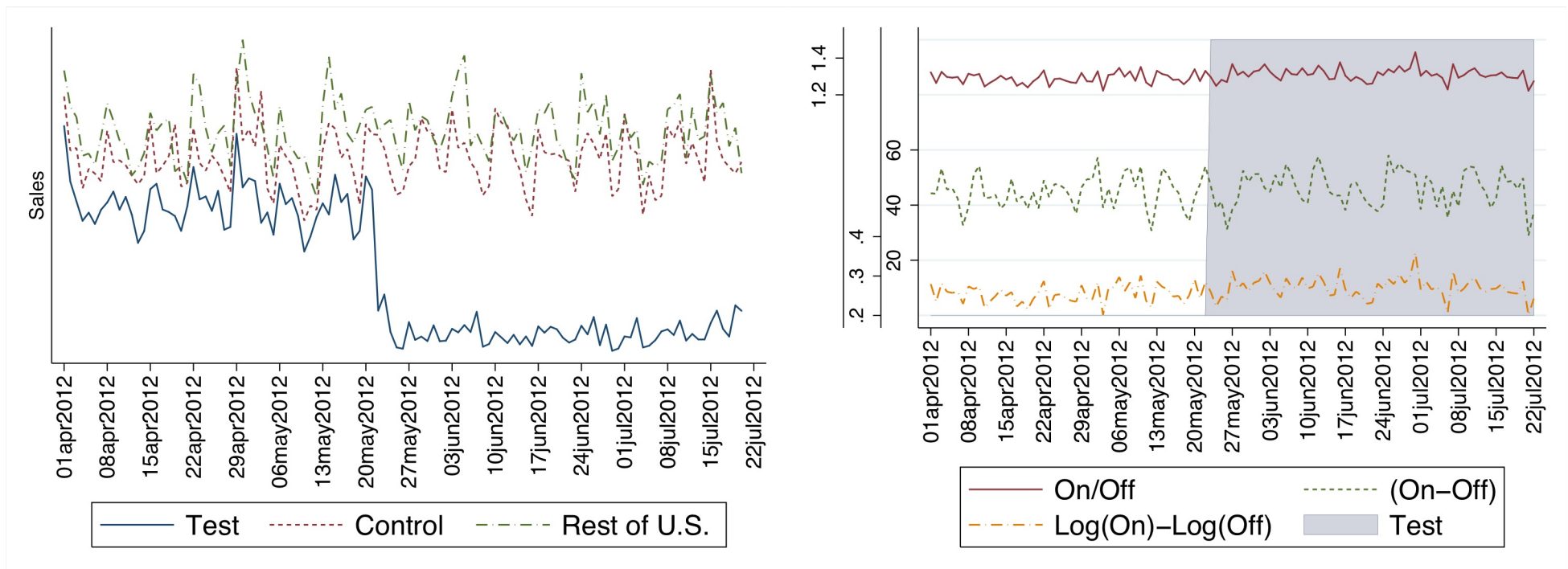
What Happens to Natural Search Clicks When Paid Ads are Turned Off?

- Paid links shutoff between July 1st and August 1st



Experiment Turns On in Second Half of Each Figure. What Happened?

Figure 3: Non-Brand Keyword Region Test



(a) Attributed Sales by Region

(b) Differences in Total Sales

Summary of results

- ROI for ad spending switches from huge and positive to -100%!
- Tens of millions of dollars in ad spending at stake...
- Very tough to change strategy

The outcome: cash savings



ebay



All



Shopping



Books



News



Videos



More

Tools

About 710,000,000 results (0.44 seconds)

<https://www.ebay.com>

eBay: Electronics, Cars, Fashion, Collectibles & More

Buy & sell electronics, cars, clothes, collectibles & more on **eBay**, the world's online marketplace. Top brands, low prices & free shipping on many items.

Search ebay.com



All Categories

Shop by department, purchase cars, fashion apparel ...

eBay Motors

Cars & Trucks - Collector Cars - Classic Cars - Boats - Boat Parts

Auto Parts & Accessories

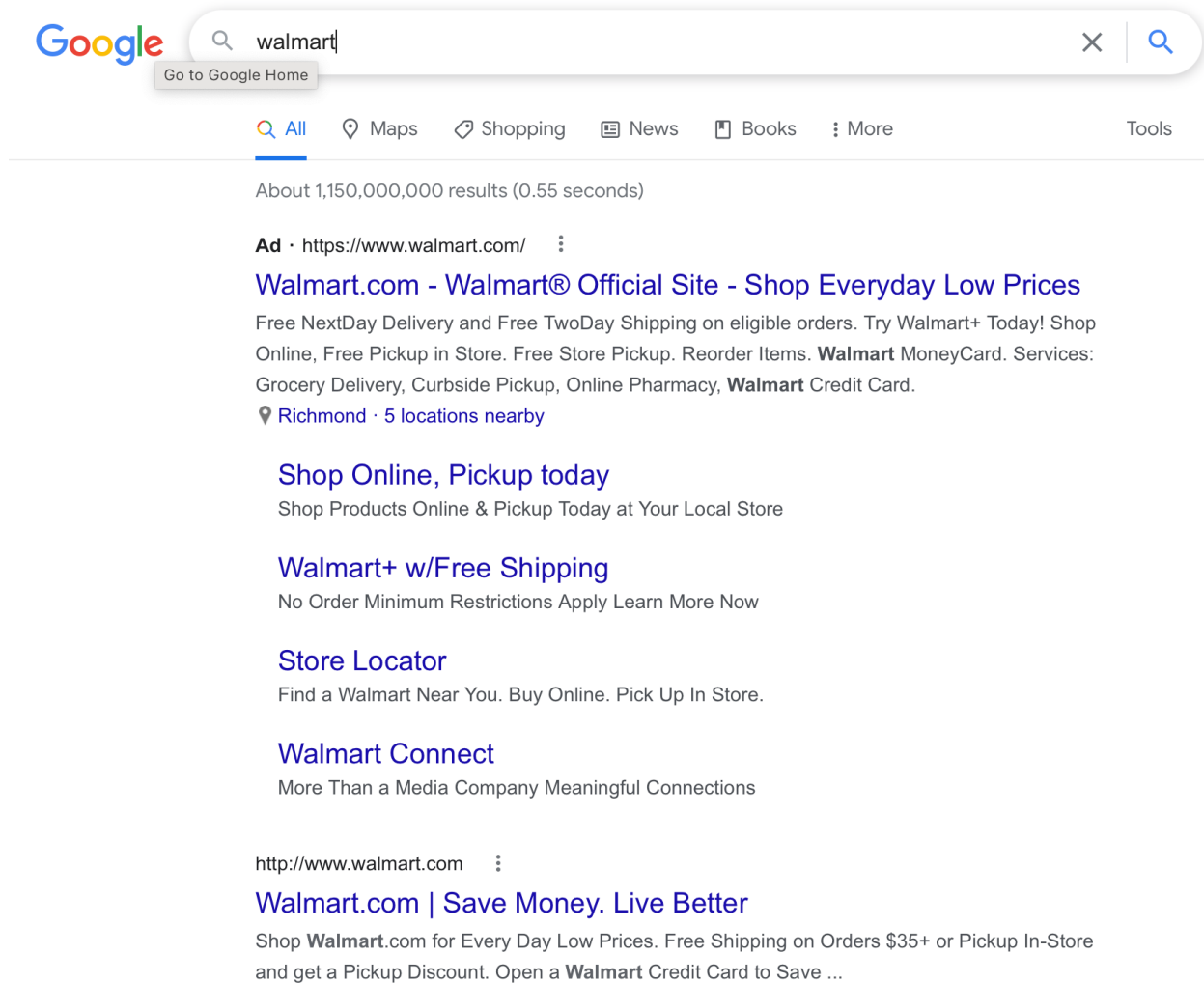
Car & Truck Parts - Motorcycle Parts - Salvage Parts Cars - Other

Cars & Trucks

Get the best deals on Cars & Trucks when you shop the ...

The outcome: cash savings

Compare Walmart... to eBay:



Google search results for 'walmart'.

Search bar: walmart

Go to Google Home

Navigation: All Maps Shopping News Books More Tools

About 1,150,000,000 results (0.55 seconds)

Ad · <https://www.walmart.com/>

Walmart.com - Walmart® Official Site - Shop Everyday Low Prices

Free NextDay Delivery and Free TwoDay Shipping on eligible orders. Try Walmart+ Today! Shop Online, Free Pickup in Store. Free Store Pickup. Reorder Items. **Walmart** MoneyCard. Services: Grocery Delivery, Curbside Pickup, Online Pharmacy, **Walmart** Credit Card.

📍 [Richmond](#) · [5 locations nearby](#)

Shop Online, Pickup today

Shop Products Online & Pickup Today at Your Local Store

Walmart+ w/Free Shipping

No Order Minimum Restrictions Apply [Learn More Now](#)

Store Locator

Find a Walmart Near You. Buy Online. Pick Up In Store.

Walmart Connect

More Than a Media Company Meaningful Connections

<http://www.walmart.com>

Walmart.com | Save Money. Live Better

Shop **Walmart.com** for Every Day Low Prices. Free Shipping on Orders \$35+ or Pickup In-Store and get a Pickup Discount. Open a **Walmart** Credit Card to Save ...

The Search World is Changing Rapidly



A new era for AI Search

May 19, 2026 | 6 min read



We're bringing our advanced model capabilities to Search with new AI features, enabling you to use agents just by asking a question. We're also introducing a new, intelligent AI-powered Search box, marking its biggest upgrade in over 25 years.

Elizabeth Reid
VP, Search

Over 70% of google searches are now “zero-click”
(because AI often gives users the answer they were looking for)

- How to earn influence and grow brand awareness without requiring a visit to your website?

Punchline

- Data + analytics present amazing opportunities
- But also amazing room for error if you don't know how to interpret what you find